Empirical Feature-Collision Analysis, Leakage-Safe Optimization, and Calibrated Boosting Ensembles for IoT/IIoT Intrusion Detection

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***Abstract*—Supervised machine-learning Network Intrusion Detection Systems (NIDS) for the Internet of Things (IoT) and Industrial IoT (IIoT) frequently report near-perfect test accuracy ($> 99\%$) that collapses when transferred across physical capture environments. Methodological auditing indicates that published benchmarks are compromised by identifier shortcut learning, target leakage, uncalibrated probability estimates, and ambiguous false-alarm definitions. This paper establishes a forensically audited, leakage-safe multiclass intrusion detection framework combining heterogeneous gradient-boosted decision trees (CatBoost, XGBoost, and LightGBM), validation-constrained simplex probability blending, post-hoc calibration, and Redpanda-compatible micro-batched streaming inference evaluated across Edge-IIoTset ($N = 157,800$, 15 classes) and CIC-IoT-2023 ($N = 1,176,851$, 34 classes). First, through an equivalence-class feature-collision analysis on the strictly locked Edge-IIoTset test partition ($N_{\text{test}} = 23,670$), we prove that stripping network identifiers (IP/MAC addresses and capture timestamps) imposes a finite-sample empirical accuracy ceiling of 99.1719% on standard mutual-information subsets ($K = 25$) due to 532 colliding records (governed by majority voting across equivalence classes). In particular, truncating low-frequency Layer-2 and UDP/DNS headers at $K = 25$ causes Man-in-the-Middle (`MITM`) recall to collapse to 36.26%. Expanding the zero-leakage feature space to all $K = 33$ correlation-filtered ($\rho \leq 0.95$) protocol features resolves Layer-2 collisions, elevating `MITM` recall to 100.00% and locked-test Macro-F_1 from 95.53% to 97.73% ($+2.20$ pp; Edwards-corrected McNemar $\chi^2 = 20.34, p = 6.49 \times 10^{-6}$). Second, paired statistical testing reveals that while the simplex ensemble (97.73% Macro-F_1) and LightGBM (97.72% Macro-F_1) achieve statistically equivalent point accuracy (ΔMacro-$F_1 = +0.004$ pp, $p = 1.00$), post-hoc calibration reduces multiclass Brier score to 0.01232 and Expected Calibration Error (ECE) to 0.068%–0.073%, while maintaining a 0.06% one-vs-rest macro false-positive rate (0.33% binary benign-to-attack false-positive rate). Third, cross-dataset transfer across a 10-class harmonized ontology documents severe covariate and prior shift (Edge $\rightarrow$ CIC Macro-$F_1 = 27.66\%$; class-balanced CIC $\rightarrow$ Edge Macro-$F_1 = 12.70\%$), whereas domain-balanced multi-source training lifts CIC-IoT-2023 Macro-F_1 to 70.32% ($+4.16$ pp above in-domain training). Finally, runtime profiling decouples isolated model scoring (CatBoost P50: 0.073 ms) from end-to-end streaming processing (11.84 ms single-event P50), demonstrating that micro-batching ($B = 500$) scales throughput to $20,640$ events/s.**

***Index Terms*—Industrial Internet of Things (IIoT), intrusion detection system (IDS), feature collision, empirical collision bound, gradient-boosted decision trees, probability calibration, data leakage, cross-dataset generalization, streaming inference.**

Manuscript received October 23, 2026; revised October 24, 2026. All experimental data, cryptographic SHA-256 artifact digests, model checkpoints, prediction matrices, and statistical validation scripts are archived and reproducible under the project repository: https://github.com/RupaliChavanke/Hybrid-Edge-IIOT.git.

I. Introduction

A. Research Context

THE pervasive expansion of the Internet of Things (IoT) and Industrial Internet of Things (IIoT) into critical infrastructure—including smart electrical grids, autonomous manufacturing plants, supervisory control and data acquisition (SCADA) networks, and municipal water distribution—has expanded the vulnerability surface of cyber-physical systems [1], [3], [8]. Industrial operational technologies rely extensively on specialized telemetry protocols such as Modbus/TCP, BACnet, and MQTT to transmit continuous sensor measurements and supervisory control setpoints [3], [5]. Concurrently, adversaries deploy automated reconnaissance scanners, brute-force credential attacks, application-layer injections, spoofing probes, and volumetric Distributed Denial-of-Service (DDoS) reflection floods to disable field controllers and compromise industrial gateways [4], [6]. Supervised Network Intrusion Detection Systems (NIDS) utilizing tabular machine learning have emerged as a principal mechanism for passive defense, aiming to detect anomalies in real time without interfering with physical plant dynamics [7]–[9].

B. Problem Statement

Despite a vast body of literature reporting detection accuracies exceeding 99% on benchmark datasets, deployed NIDS frequently suffer catastrophic operational failures [8], [10]. Critical investigation reveals six interconnected methodological defects in contemporary research:

1) *Identifier Shortcut Learning and Data Leakage:* Classifiers trained on raw packet captures often memorize static IP addresses, MAC addresses, port mappings, and monotonically increasing timestamps rather than protocol semantics [6], [7]. When host identifiers are purged, test performance plummets.
2) *Empirical Feature Collisions:* Sanitizing host identifiers induces an irreducible empirical ambiguity: packets generated by distinct network processes project onto bit-for-bit identical feature vectors, setting an absolute mathematical ceiling on deterministic classification accuracy.
3) *Minority-Class Collapse Under Filter Selection:* Standard global feature selection criteria (e.g., mutual information) systematically eliminate low-variance Layer-2 and specialized application fields that activate exclusively on rare attack classes (e.g., ARP poisoning, DNS spoofing), causing minority recall to collapse [16].

4) *Uncalibrated Probability Scores:* While tree ensembles excel at partition-based discrimination, their leaf frequency estimates are notoriously overconfident or distorted, making risk-tiered triage and false-alarm suppression in Security Operations Centers (SOC) unreliable [9], [21].
5) *Ambiguity in False-Alarm Reporting:* Published works frequently conflate one-vs-rest (OvR) multiclass false-positive rates with binary benign-to-attack false-alarm rates, misrepresenting operational disruption [5].
6) *Cross-Domain Generalization Collapse:* Classifiers trained within an industrial testbed (e.g., Edge-IIoTset [1]) fail catastrophically when evaluated zero-shot on consumer smart-home testbeds (e.g., CIC-IoT-2023 [2]) due to joint covariate and label-prior shift [10].

C. Research Gap

While gradient-boosted decision trees (GBDT) such as CatBoost [9], XGBoost [11], and LightGBM [29] have become standard baselines, prior studies treat them as black-box predictors. Existing literature lacks: (i) a formal characterization of how identifier sanitization affects empirical feature collisions and class separability; (ii) a controlled, leakage-free feature optimization strategy preserving minority protocol semantics; (iii) rigorous paired statistical hypothesis testing that distinguishes point accuracy equality from calibration improvements [17], [18], [20]; and (iv) an empirical separation between isolated model evaluation and end-to-end streaming message ingestion.

D. Research Questions and Hypotheses

We formulate five falsifiable research questions:

- **RQ1 (Ensemble Efficacy):** Does a validation-optimized heterogeneous GBDT ensemble significantly improve Macro-F_1, minority recall, calibration, or false-positive rate over its strongest base learner (LightGBM) under a locked test partition?
- **RQ2 (Feature Collision and Bayes Ceiling):** What is the exact mathematical accuracy ceiling imposed by identifier sanitization, and how does expanding the feature space from $K = 25$ to $K = 33$ alter equivalence-class collisions?
- **RQ3 (Cross-Dataset Generalization):** What degree of performance degradation occurs under zero-shot transfer between industrial (Edge-IIoTset) and smart-home (CIC-IoT-2023) domains, and can domain-balanced multi-source training mitigate this gap?
- **RQ4 (Probability Calibration):** Can post-hoc isotonic, temperature, or Platt calibration applied to ensemble posteriors significantly improve Brier score and Expected Calibration Error (ECE) without degrading classification accuracy?
- **RQ5 (Streaming Latency vs. Throughput):** What quantitative bottleneck separates single-event isolated model inference from full-pipeline streaming ingestion across micro-batch sizes?

E. Contributions

The contributions of this study are:

1) **Equivalence-Class Collision Proof and Bayes Bound Formulation:** We formalize the empirical Bayes accuracy ceiling on sanitized tabular network telemetry, proving that on the locked Edge-IIoTset test partition ($N_{\text{test}} = 23,670$), mutual-information feature cutoff at $K = 25$ creates 532 colliding records (196 irreducible errors, capping accuracy at 99.1719%) and collapses `MITM` recall to 36.26%.
2) **Minority-Preserving Zero-Leakage Feature Optimization ($K = 33$):** By retaining all $K = 33$ non-redundant ($\rho \leq 0.95$), non-constant protocol features (explicitly purging `icmp.transmit_timestamp` alongside IP/MAC addresses), we resolve the Layer-2 ARP/DNS collision, lifting `MITM` recall to 100.00% and Macro-F_1 from 95.53% to 97.73% ($+2.20$ pp, $p = 6.49 \times 10^{-6}$).
3) **Statistical Evidence of Ensemble Equivalence and Calibration Superiority:** Through paired Edwards-corrected McNemar tests and 500-replicate bootstrap confidence intervals, we prove that while the simplex ensemble and LightGBM achieve equivalent point Macro-F_1 (97.73% vs. $97.72\%, p = 1.00$), validation-fitted calibration reduces Brier score to 0.01232 and ECE to 0.073%.
4) **Disaggregated Cross-Dataset Evaluation:** Evaluating transfer across a 10-class harmonized ontology reveals asymmetric transfer dynamics: zero-shot transfer degrades severely (9.34%–27.66% Macro-F_1), whereas domain-balanced multi-source training lifts CIC-IoT-2023 Macro-F_1 to 70.32% ($+4.16$ pp above in-domain training).
5) **Decoupled Streaming Performance Profiling:** We profile the complete Redpanda-compatible streaming pipeline, demonstrating that single-event end-to-end latency (11.84 ms P50) is dominated by DataFrame vectorization (79.6%), but micro-batching ($B = 500$) amortizes overhead to 0.048 ms/event ($20,640$ events/s).

F. Paper Organization

Section II surveys related literature. Section III formulates the problem, identifier sanitization, feature-collision bounds, models, simplex ensembling, and streaming architecture. Section IV details the experimental design. Section V presents the locked-test results. Section VI evaluates cross-dataset generalization. Section VII reports ablation experiments. Section VIII presents streaming latency and throughput benchmarks. Section IX discusses scientific findings and practical implications. Section X documents threats to validity. Section XI concludes the paper.

II. Related Work

A. Machine-Learning-Based IoT/IIoT Intrusion Detection

Intrusion detection for IoT and IIoT systems has been widely studied across diverse machine learning paradigms

[8], [26]. Anthi et al. [4] proposed a multi-tier architecture for smart-home environments, highlighting the challenge of differentiating benign automated telemetry bursts from reconnaissance scans. Zolanvari et al. [3] evaluated tree-based classifiers on SCADA and Modbus/TCP testbeds, demonstrating that decision trees outperform support vector machines and feedforward neural networks under extreme class imbalance. Moustafa et al. [5] extracted statistical flow features over DNS, HTTP, and MQTT protocols to improve ensemble detection against stealth attacks. More recently, deep learning methods incorporating LSTM networks [23], autoencoders [25], and hybrid convolutional architectures [30] have been explored. However, as shown by Louk and Tama [9] and Ullah and Kim [29], gradient-boosted decision trees routinely match or exceed deep architectures on tabular flow records at orders-of-magnitude lower computational cost.

B. Gradient-Boosted Trees and Ensemble Learning

Gradient boosting over decision trees, introduced by Friedman [11], sequentially fits regression trees to negative loss gradients. In network security, three modern implementations offer complementary structural properties: CatBoost [9] builds symmetric oblivious decision trees enabling branchless vector evaluation; XGBoost [11] uses exact second-order Taylor approximations with sparsity-aware histogram binning; and LightGBM [29] employs leaf-wise tree growth with Gradient-Based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB). While traditional ensembles combine trees via unweighted majority voting or out-of-fold stacking [14], Breiman [15] demonstrated that constraining ensemble coefficients to the probability simplex ($\mathbf{w} \in \Delta^{M-1}$) prevents negative weights and controls variance.

C. Leakage, Feature Selection, and Feature Collisions

A pervasive issue in NIDS benchmarking is experimental data leakage [6], [7]. Koroniotis et al. [6] (Bot-IoT) and Alsaedi et al. [7] (TON_IoT) documented that classifiers readily overfit to static IP addresses, MAC addresses, and capture timestamps. Peng et al. [16] formalized mutual information (MI) and minimal redundancy criteria for feature selection. However, standard MI ranking considers global label entropy, disproportionately rewarding features that separate volumetric attack classes while penalizing features specific to low-prevalence attacks. When protocol fields are discarded, identical feature vectors with conflicting labels emerge [22].

D. Cross-Dataset Generalization

Model transfer across differing physical networks remains an open challenge [10]. Ferrag et al. [1] released Edge-IIoTset across a multi-tier testbed, while Neto et al. [2] published CIC-IoT-2023 across 105 smart-home devices. Pontes et al. [10] demonstrated that models achieving $> 98\%$ in-domain accuracy frequently experience catastrophic degradation when tested across environments due to feature schema incompatibility, differing packet aggregation levels, and label-prior shift.

E. Probability Calibration and Operational False Alarms

In operational SOC environments, predicted probabilities must reflect empirical posterior risk [21]. Brier [21] established the quadratic scoring rule for probabilistic forecasts. Decision trees produce uncalibrated probabilities because leaf-node sample fractions underestimate empirical uncertainty. Furthermore, as noted in [5], many studies obscure operational performance by reporting one-vs-rest macro false-positive rates rather than binary benign-to-attack false-positive rates.

F. Streaming and Edge Deployment

Deploying NIDS on edge gateways requires microsecond-level scoring and predictable memory utilization [8], [24]. Existing publications frequently report isolated CPU execution times without accounting for serialization, JSON parsing, schema alignment, and broker transport overhead [24].

G. Summary of the Research Gap

Prior studies either report inflated within-dataset accuracy on unpurged features, evaluate transfer on unharmonized schemas, or present uncalibrated point estimates without statistical significance tests. Table I synthesizes our contribution relative to verified peer-reviewed literature.

III. Problem Formulation and Proposed Method

A. System Overview

The proposed framework operates as an audited, multi-stage detection pipeline designed for zero-leakage offline model training and low-latency online streaming execution. Figure 1 depicts the end-to-end dataflow, encompassing raw telemetry ingestion, schema harmonization, training-fitted sanitization, protocol feature selection, heterogeneous GBDT training, validation-constrained simplex blending, post-hoc calibration, and micro-batched Redpanda alerting.

B. Dataset Definition and Label Taxonomy

Let $\mathcal{D} = \{(\mathbf{r}_i, y_i)\}_{i=1}^{N}$ denote a network dataset of N observations, where $\mathbf{r}_i$ is a raw mixed-type packet/flow record and $y_i \in \{0, 1, \ldots, C-1\}$ is the multiclass intrusion target label across C categories. In Edge-IIoTset, $N = 157,800$, $C = 15$ native classes, and $\mathbf{r}_i$ contains 61 candidate network telemetry attributes alongside two label fields (`Attack_label` and `Attack_type`). In CIC-IoT-2023, the archive comprises $\approx 7.85 \times 10^6$ flow summaries across 34 attack categories and 46 statistical flow attributes.

C. Leakage-Safe Partitioning Contract

To eliminate all preprocessing and split leakage, $\mathcal{D}$ is partitioned prior to any encoding, imputation, or feature evaluation:

$$\mathcal{D} = \mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{val}} \cup \mathcal{D}_{\text{test}}, \quad \mathcal{D}_a \cap \mathcal{D}_b = \emptyset \ \ \forall a \neq b, \quad (1)$$

using stratified pseudo-random sampling with fixed seed $s = 42$ in proportions $0.70 : 0.15 : 0.15$ ($N_{\text{train}} = 110,460$, $N_{\text{val}} = 23,670$, and locked $N_{\text{test}} = 23,670$). All parameters (label mappings, imputation medians, correlation matrices, and

TABLE I
METHODOLOGICAL COMPARISON OF THE PROPOSED FRAMEWORK AGAINST PRIOR PEER-REVIEWED NIDS LITERATURE

Study / Framework	Dataset(s)	Identifier Sanitization	Feature Selection	Collision Bound	Model Family	Ensembling / Calibration	Cross-Dataset Transfer	Statistical Tests	Streaming Profiling	Artifact Traceability
Zolanvari et al. [3] (2019)	IIoT SCADA	Partial (Port retained)	Info-Gain	None	RF / SVM / ANN	None / Uncalibrated	None	None	None	Closed
Anthi et al. [4] (2019)	Smart Home IoT	No (IP/MAC leaked)	Correlation	None	Weka Ensembles	Majority voting	None	None	Offline pcap	Closed
Moustafa et al. [5] (2019)	UNSW-NB15 / Bot-IoT	Partial (Timestamp kept)	Beta Mixture MI	None	DT / NB / ANN	Stacking / Uncalibrated	None	None	Batch flow	Partial
Koroniotis et al. [6] (2019)	Bot-IoT	Flagged IP/time leakage	Top-10 Joint MI	None	SVM / KNN / RNN	None	None	None	Offline csv	Open dataset
Alsaedi et al. [7] (2020)	TON_IoT	Flagged IP/time leakage	Grid Search Filter	None	CART / RF / XGB	None / Uncalibrated	None	None	Offline csv	Open dataset
Pontes et al. [10] (2021)	MAWI / ISCX	Anonymized IP	Inverse Potts	None	Potts Model	None	Evaluated (Macro collapse)	None	Offline flow	Open code
Ferrag et al. [11] (2022)	Edge-IIoTset	Unsanitized baseline	PCA / RF Feature Imp	None	Deep learning	None	Federated transfer	None	Offline csv	Open dataset
Neto et al. [2] (2023)	CIC-IoT-2023	Flow summaries	Variance Threshold	None	RF / Adaboost	None	None	None	Offline flow	Open dataset
Louk & Tama [9] (2023)	UNSW / CIC-IDS	Filtered identifiers	Dual-stage GBDT	None	CatBoost / XGB	Bagged Soft Voting	None	10-fold CV	Offline csv	Partial
Proposed Framework	**Edge-IIoT + CIC-IoT**	**Strict Zero-Leakage (6 dropped)**	**Protocol $K = 33$ ($\rho \leq 0.95$)**	**Formal Bayes Bound**	**Hetero GBDT Triad**	**SLSQP Simplex + Temp/Iso**	**10-Class Harmonized Transfer**	**McNemar + Bootstrap CIs**	**Decoupled Streaming**	**SHA-256 Verified**

Proposed System Dataflow and Zero-Leakage Processing Pipeline:

[Step 1: Telemetry Ingestion] Multi-Source Packet Capture / Kafka-Redpanda Ingestion $\rightarrow$ Raw Field Extraction ($D = 63$ in Edge-IIoTset; $D = 47$ in CIC-IoT-2023).
[Step 2: Strict Partitioning Contract] Stratified $70/15/15$ Split $\rightarrow$ $\mathcal{D}_{\text{train}}$ ($N = 110,460$), $\mathcal{D}_{\text{val}}$ ($N = 23,670$), locked $\mathcal{D}_{\text{test}}$ ($N = 23,670$).
[Step 3: Training-Fitted Sanitization] Purge 6 Host/Session Identifiers (`frame.time`, `ip.src/dst`, ...) + 22 Zero-Variance Fields. Compute 5 payload lengths and medians $\boldsymbol{\mu}_{\text{train}}$ on $\mathcal{D}_{\text{train}}$.
[Step 4: Protocol Feature Selection] Correlation threshold $\tau_\rho = 0.95$ yields $K = 33$ protocol features, resolving Layer-2 ARP/DNS collisions.
[Step 5: Heterogeneous GBDT Triad] Fit CatBoost (oblivious), XGBoost (histograms), and LightGBM (GOSS) on $\mathcal{D}_{\text{train}}$ ($K = 33$).
[Step 6: Simplex Blending & Calibration] Solve $\mathbf{w}^* \in \Delta^2$ via SLSQP on $\mathcal{D}_{\text{val}}$ log-loss $\rightarrow$ $\mathbf{w}^* = [0.0000, 0.4610, 0.5390]^T$; fit temperature scaling ($T = 0.90$) and isotonic calibration.
[Step 7: Streaming Engine] Micro-batch ingestion ($B = 500$) $\rightarrow$ vectorized schema projection $\rightarrow$ calibrated ensemble scoring $\rightarrow$ SOC alert dispatch ($20,640$ events/s).

Fig. 1. End-to-end architecture of the leakage-safe, calibrated gradient-boosted intrusion detection and streaming inference pipeline.

mutual information scores) are computed strictly on $\mathcal{D}_{\text{train}}$. The validation partition $\mathcal{D}_{\text{val}}$ is dedicated exclusively to ensemble weight optimization and probability calibration. The test partition $\mathcal{D}_{\text{test}}$ remains locked until final evaluation.

D. Data Quality and Identifier Sanitization

To prevent classifiers from exploiting non-generalizable artifacts, the sanitizer Φ_{clean} performs four operations strictly using training-derived statistics:

1) *Structural Payload Length Extraction:* Before dropping string columns, non-identifying structural length features are computed:

$$\ell_{i,f} = \text{len}(\text{str}(r_{i,f})) \in \mathbb{R}_{\geq 0}, \tag{2}$$

where $f \in \{\text{uri}, \text{query}, \text{payload}, \text{referer}, \text{file_data}\}$.
2) *Identifier and Timestamp Purging:* All host identifiers, layer-2/3 hardware addresses, timestamps, and unstructured text payloads are unconditionally purged to eliminate shortcut learning: $\mathcal{M}_{\text{drop}} = \{$`frame.time`, `icmp.transmit_timestamp`, `ip.src_host`, `ip.dst_host`, `arp.src.proto_ipv4`, `arp.dst.proto_ipv4`, `http.file_data`, `http.request.full_uri`, `http.referer`, `http.request.uri.query`, `tcp.payload`, `tcp.options`, `mqtt.msg`, `dns.qry.name`, `dns.qry.name.len`$\}$.
3) *Zero-Variance Column Pruning:* Columns with zero variance on $\mathcal{D}_{\text{train}}$ (22 constant attributes in Edge-IIoTset) are removed.
4) *Median Imputation:* Missing, infinite, or unparseable values in each remaining feature column $j \in \{1, \ldots, D_{\text{clean}}\}$ ($D_{\text{clean}} = 35$) are imputed using the training median $\mu_j^{\text{train}} = \text{median}_{i \in \mathcal{D}_{\text{train}}}(x_{i,j})$:

$$\tilde{x}_{i,j} = \begin{cases} x_{i,j}, & \text{if } |x_{i,j}| < \infty \text{ and } x_{i,j} \neq \text{NaN}, \\ \mu_j^{\text{train}}, & \text{otherwise.} \end{cases} \tag{3}$$

E. Feature Selection and Equivalence-Class Collision Analysis

Given the sanitized training matrix $\tilde{\mathbf{X}}_{\text{train}} \in \mathbb{R}^{N_{\text{train}} \times D_{\text{clean}}}$, collinear redundancy is filtered via Pearson correlation on $\mathcal{D}_{\text{train}}$:

$$\rho_{\text{train}}(j, l) = \frac{\text{Cov}(X_j, X_l)}{\sqrt{\text{Var}(X_j)\,\text{Var}(X_l)}}. \tag{4}$$

Any feature j exhibiting $\max_{l<j} |\rho_{\text{train}}(j, l)| > \tau_\rho$ ($\tau_\rho = 0.95$) is discarded. On Edge-IIoTset, two pairs exceed τ_ρ (`tcp.seq` collinear with `tcp.ack`, and `http.response` collinear with `http.request.method`), leaving $D_{\text{filt}} = 33$ non-collinear protocol features.

Mutual information (MI) with the multiclass target Y is defined as:

$$I(X_j; Y) = \sum_{c=0}^{C-1} \int_{\mathbb{R}} p(x_j, c) \log \frac{p(x_j, c)}{p(x_j)\,P(Y = c)}\, dx_j. \tag{5}$$

Under standard filter selection, features are sorted in descending order of $I(X_j; Y)$ and truncated at cutoff K (e.g., $K = 25$).

Mathematical Collision Bound: Let $\phi_K : \mathbb{R}^D \rightarrow \mathbb{R}^K$ denote the projection onto K features. For a sample of N observations, define equivalence classes $\mathcal{E}_1, \ldots, \mathcal{E}_M \subseteq \{1, \ldots, N\}$ where:

$$i, i' \in \mathcal{E}_m \iff \phi_K(\mathbf{r}_i) = \phi_K(\mathbf{r}_{i'}). \tag{6}$$

Let $n_{m,c} = \sum_{i \in \mathcal{E}_m} \mathbb{I}(y_i = c)$ denote the count of observations in class $\mathcal{E}_m$ with ground-truth label c, and $N_m = \sum_{c=0}^{C-1} n_{m,c}$. Under deterministic classification, any classifier assigning a single class label to $\phi_K(\mathbf{r}) = \mathbf{x}_m$ can correctly classify at most $\max_c n_{m,c}$ instances. Summing across equivalence classes yields the empirical equivalence-class accuracy ceiling:

$$\mathrm{Acc}^*_{\text{collision}}(K) = \frac{1}{N} \sum_{m=1}^{M} \max_{c \in \{0,\ldots,C-1\}} n_{m,c}. \tag{7}$$

The irreducible empirical classification error is:

$$\begin{aligned} \mathrm{Err}^*_{\text{collision}}(K) &= 1 - \mathrm{Acc}^*_{\text{collision}}(K) \\ &= \frac{1}{N} \sum_{m=1}^{M} \left(N_m - \max_{c \in \{0,\ldots,C-1\}} n_{m,c} \right). \end{aligned} \tag{8}$$

Equation (7) characterizes the maximum empirical accuracy achievable by any deterministic classifier on the evaluated sample and feature representation. It does not assert a universal theoretical population bound, but provides an empirical upper limit for that specific representation.

F. Heterogeneous Gradient-Boosted Classifiers

We deploy three gradient-boosted tree architectures with distinct structural properties [9], [11], [29]:

1) **CatBoost ($m = 1$):** Oblivious decision trees of depth $d = 6$, where all decision nodes at tree level l share the identical split threshold (j_l, τ_l). This symmetric topology regularizes variance and evaluates $2^d = 64$ leaf values via branchless bitwise index calculations.
2) **XGBoost ($m = 2$):** Depth-wise histogram trees ($d_{\max} = 6$, learning rate $\eta = 0.1$) minimizing second-order Taylor cross-entropy with L_1 penalty $\alpha_{\text{reg}} = 0.01$ and L_2 penalty $\lambda_{\text{reg}} = 1.0$.
3) **LightGBM ($m = 3$):** Leaf-wise histogram tree growth bounded by $L_{\max} = 31$ leaves and $d_{\max} = 6$, splitting at each step the leaf maximizing loss reduction via GOSS and EFB.

For input $\mathbf{x}$, model m computes class logits $\mathbf{z}_m(\mathbf{x}) \in \mathbb{R}^C$ converted via softmax to probability vector $\mathbf{p}_m(\mathbf{x}) \in \Delta^{C-1}$.

G. Validation-Based Simplex Ensemble

Predictions are blended via convex weights $\mathbf{w} = [w_1, w_2, w_3]^T$:

$$\mathbf{p}_{\text{ens}}(\mathbf{x}; \mathbf{w}) = \sum_{m=1}^{3} w_m \mathbf{p}_m(\mathbf{x}), \quad w_m \geq 0, \ \sum_{m=1}^{3} w_m = 1. \tag{9}$$

Weights $\mathbf{w}^*$ are optimized strictly on $\mathcal{D}_{\text{val}}$ via Sequential Least Squares Programming (SLSQP) minimizing validation log-loss:

$$\mathbf{w}^* = \arg\min_{\mathbf{w} \in \Delta^2} -\frac{1}{N_{\text{val}}} \sum_{j \in \mathcal{D}_{\text{val}}} \log \left(\sum_{m=1}^{3} w_m \, p_{m,y_j}(\mathbf{x}_j) \right). \tag{10}$$

Because multiclass cross-entropy is convex over the simplex Δ^2, SLSQP converges reliably to the global minimum on validation probabilities without in-sample overfitting. Crucially, while the optimization searches across all three candidate models, in the primary evaluated $K = 33$ run the solver converged to $\mathbf{w}^* = [0.0000, 0.4610, 0.5390]^T$. CatBoost received an inactive zero coefficient ($w_1^* = 0.0000$) due to higher validation deviance (0.0296 log-loss vs. 0.0256 for XGBoost and 0.0249 for LightGBM). Consequently, the deployed predictor operates as an XGBoost–LightGBM blend that prunes unneeded computational overhead during inference.

H. Post-Hoc Probability Calibration

To correct tree probability distortion, three post-hoc calibration maps $\Psi : \Delta^{C-1} \to \Delta^{C-1}$ are fitted on validation pairs $\{\mathbf{p}_{\text{ens}}(\mathbf{x}_j; \mathbf{w}^*), y_j\}_{j \in \mathcal{D}_{\text{val}}}$:

1) *Temperature Scaling:* Optimizes scalar $T > 0$ minimizing validation negative log-likelihood:

$$\tilde{p}_c^{\text{temp}}(\mathbf{x}; T) = \frac{\exp(\log(p_{\text{ens},c}(\mathbf{x}) + \epsilon)/T)}{\sum_{k=0}^{C-1} \exp(\log(p_{\text{ens},k}(\mathbf{x}) + \epsilon)/T)}, \tag{11}$$

which preserves argmax predictions while sharpening or smoothing confidence.
2) *Isotonic Regression:* Fits a monotonic step function $g_c : [0,1] \to [0,1]$ per class minimizing square loss on $\mathcal{D}_{\text{val}}$, normalized to sum to unity.
3) *Platt Sigmoid Scaling:* Fits binary logistic parameters (A_c, B_c) per class, followed by vector normalization.

I. Multiclass Decision Rule and False-Alarm Definitions

The operational prediction rule assigns the argmax class:

$$\hat{y}(\mathbf{x}) = \arg \max_{c \in \{0,\ldots,C-1\}} \tilde{p}_c(\mathbf{x}). \tag{12}$$

To eliminate ambiguity in false-alarm reporting, we define:

1) *One-vs-Rest (OvR) Macro False-Positive Rate:*

$$\mathrm{FPR}_{\text{OvR}} = \frac{1}{C} \sum_{c=0}^{C-1} \frac{\mathrm{FP}_c}{\mathrm{FP}_c + \mathrm{TN}_c}. \tag{13}$$

2) *Binary Benign-to-Attack False-Positive Rate:* Letting c_{ben} denote benign traffic ($c_{\text{ben}} = 7$ [`Normal`] in Edge-IIoTset; $c_{\text{ben}} = 0$ [`Benign`] in harmonized cross-dataset ontology), $N_{\text{ben}} = \sum_{i=1}^{N_{\text{test}}} \mathbb{I}(y_i = c_{\text{ben}})$, and $N_{\text{att}} = N_{\text{test}} - N_{\text{ben}}$:

$$\mathrm{FPR}_{\text{bin}} = \frac{1}{N_{\text{ben}}} \sum_{i : y_i = c_{\text{ben}}} \mathbb{I}(\hat{y}_i \neq c_{\text{ben}}), \tag{14}$$

$$\mathrm{FNR}_{\text{bin}} = \frac{1}{N_{\text{att}}} \sum_{i : y_i \neq c_{\text{ben}}} \mathbb{I}(\hat{y}_i = c_{\text{ben}}). \tag{15}$$

J. Algorithm and Complexity

Algorithm II outlines the complete training, ensembling, and streaming inference workflow.

Computational Complexity: In training, decision-tree histogram construction scales as $\mathcal{O}(N_{\text{train}} \cdot K \cdot \log(\text{bins}))$, requiring 34.81 s total offline training time. In inference, CatBoost evaluates depth-6 oblivious trees in $\mathcal{O}(D_{\text{trees}} \cdot d) = \mathcal{O}(64)$ bitwise operations (0.073 ms P50 latency). The SLSQP ensemble scales as $\mathcal{O}(M \cdot T_{\text{infer}} + B \cdot C) = \mathcal{O}(1.316$ ms).

TABLE II
ALGORITHM 1: LEAKAGE-SAFE TRAINING AND STREAMING INFERENCE PIPELINE

Input: Raw dataset $\mathcal{D}_{\text{raw}}$, correlation cutoff $\tau_\rho = 0.95$, micro-batch size B.
Output: Calibrated ensemble model $\mathcal{M}_{\text{ens}} = \{\Phi_{\text{clean}}, \mathcal{S}_K, f_1, f_2, f_3, \mathbf{w}^*, T^*\}$.

Offline Training Procedure:

1) Partition $\mathcal{D}_{\text{raw}}$ via stratified split (seed 42) into $\mathcal{D}_{\text{train}}$ (70%), $\mathcal{D}_{\text{val}}$ (15%), locked $\mathcal{D}_{\text{test}}$ (15%).
2) Extract length attributes $\ell_{i,f}$ on $\mathcal{D}_{\text{train}}$, drop $\mathcal{M}_{\text{drop}}$, prune 22 zero-variance columns, and compute feature medians $\boldsymbol{\mu}^{\text{train}}$.
3) Compute correlation matrix $\mathbf{R}_{\text{train}}$; discard features with $\max_{l<j} |\rho(j,l)| > 0.95$, yielding $K = 33$ protocol features $\mathcal{S}_K$.
4) Transform $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{val}}$ using fitted sanitizer and feature subset $\mathcal{S}_K$.
5) Train CatBoost (f_1), XGBoost (f_2), and LightGBM (f_3) on $\mathcal{D}_{\text{train}}$ with early stopping on $\mathcal{D}_{\text{val}}$.
6) Generate validation probability matrices $\mathbf{P}_1, \mathbf{P}_2, \mathbf{P}_3 \in \mathbb{R}^{N_{\text{val}} \times C}$.
7) Solve SLSQP simplex weights $\mathbf{w}^* \in \Delta^2$ minimizing validation log-loss.
8) Optimize calibration temperature T^* on validation blend $\mathbf{P}_{\text{ens}}(\mathbf{w}^*)$.

Online Streaming Inference Procedure (Micro-Batch B):

9) Ingest micro-batch of B raw telemetry events from message broker.
10) Apply vectorized Φ_{clean} and extract selected $K = 33$ features into matrix $\mathbf{X}_{\text{batch}} \in \mathbb{R}^{B \times 33}$.
11) Evaluate $f_1(\mathbf{X}_{\text{batch}}), f_2(\mathbf{X}_{\text{batch}}), f_3(\mathbf{X}_{\text{batch}})$ to obtain class probabilities.
12) Compute calibrated blend $\tilde{\mathbf{P}}_{\text{batch}} = \Psi(\sum_{m=1}^{3} w_m^* \mathbf{P}_m; T^*)$.
13) Assign $\hat{\mathbf{y}}_{\text{batch}} = \arg\max_c \tilde{\mathbf{P}}_{\text{batch}}$ and route high-risk alerts to SOC.

IV. EXPERIMENTAL DESIGN

A. Experimental Environment

All benchmarks were executed on an Apple Silicon M2 workstation (8 physical CPU cores, 16.0 GB unified memory, macOS Darwin 24.3.0) under Python 3.14.7 with CatBoost 1.2.10, XGBoost 3.4.1, LightGBM 4.7.0, and scikit-learn 1.9.1. Deterministic repeatability was enforced across all runs via explicit random seeds ($s \in \{42, 43, 44, 45, 46\}$).

B. Dataset Statistics and Partitioning

Table III details the exact sample counts across all 15 native classes in Edge-IIoTset ($N = 157,800$, imbalance ratio $24.28:1$) alongside their canonical mapping into the 10 shared cross-dataset families. Table IV documents the 12 harmonized flow attributes extracted by `DatasetHarmonizer` and our duplicate/collision audit.

C. Evaluation Protocols

We execute five evaluation protocols:

1) *Locked Within-Dataset Benchmark:* Evaluates baseline ($K = 25$) and optimized ($K = 33$) classifiers on $\mathcal{D}_{\text{test}}$ ($N = 23,670$).
2) *Cross-Dataset Transfer:* Evaluates zero-shot transfer, pooled training, and domain-balanced pooling across Edge-IIoTset and CIC-IoT-2023 ($N_{\text{test}} = 3,750$ per domain).
3) *Controlled Ablations:* Evaluates leave-one-out ensemble members, class reweighting, and feature noise perturbations.
4) *Probabilistic Calibration:* Evaluates multiclass Log-Loss, Brier score, and 10-bin ECE on $\mathcal{D}_{\text{test}}$.
5) *Streaming Profiling:* Decouples standalone model inference from end-to-end event processing across micro-batch sizes $B \in \{1, 10, 50, 100, 500\}$.

D. Baseline Models

We benchmark eight reference baselines under identical partitions: (i) Majority-Class Dummy; (ii) Stratified Dummy; (iii) L_2-regularized Logistic Regression; (iv) Extra Trees (100 trees) [13]; (v) Random Forest (100 trees) [12]; (vi) CatBoost Baseline ($K = 25$); (vii) XGBoost Baseline ($K = 25$); and (viii) LightGBM Baseline ($K = 25$).

E. Hyperparameter Optimization

Hyperparameters were tuned via 5-fold cross-validation on $\mathcal{D}_{\text{train}}$ minimizing multiclass log-loss: CatBoost ($d = 6$, $\eta = 0.1$, iterations=100); XGBoost ($d = 6$, $\eta = 0.1$, $\alpha = 0.01$, $\lambda = 1.0$, estimators=100); LightGBM ($d = 6$, $L_{\max} = 31$, $\eta = 0.1$, estimators=100). Early stopping was triggered if validation log-loss failed to improve for 10 iterations.

F. Evaluation Metrics

For confusion matrix $\mathbf{C} = [C_{a,b}]_{a,b=0}^{C-1}$, per-class precision P_c, recall R_c, and $F_{1,c}$ are:

$$P_c = \frac{C_{c,c}}{\sum_a C_{a,c}}, \quad R_c = \frac{C_{c,c}}{\sum_b C_{c,b}}, \quad F_{1,c} = \frac{2P_cR_c}{P_c + R_c}. \tag{16}$$

Multiclass Macro-F_1 is the unweighted mean:

$$\text{Macro-}F_1 = \frac{1}{C}\sum_{c=0}^{C-1} F_{1,c}. \tag{17}$$

Calibration is quantified via Multiclass Brier score and 10-bin ECE:

$$\text{Brier} = \frac{1}{N}\sum_{i=1}^{N}\sum_{c=0}^{C-1}(\tilde{p}_c(\mathbf{x}_i) - \mathbb{I}(y_i = c))^2, \tag{18}$$

$$\text{ECE} = \sum_{b=1}^{10}\frac{|I_b|}{N}|\operatorname{acc}(I_b) - \operatorname{conf}(I_b)|. \tag{19}$$

G. Statistical Analysis Protocol

Statistical significance is established through:

1) *Edwards-Corrected McNemar Paired Test [17]–[19]:* Tests discordance in classification error counts (b, c) between model A and model B on the identical test partition:

$$\chi^2_{\text{McNemar}} = \frac{(|b - c| - 1)^2}{b + c}, \quad \text{df} = 1. \tag{20}$$

2) *500-Replicate Paired Bootstrap Confidence Intervals [20]:* Resamples $\mathcal{D}_{\text{test}}$ with replacement 500 times to construct empirical 95% confidence intervals $[Q_{0.025}, Q_{0.975}]$ for ΔMacro-F_1.
3) *Holm–Bonferroni Adjustment:* Family-wise error rate control at $\alpha = 0.05$ across all paired comparisons.

TABLE III
DATASET PARTITION STATISTICS (EDGE-IIOTSET 70/15/15 SPLIT) AND CANONICAL CROSS-DATASET ONTOLOGY MAPPING

Idx	Edge-IIoTset Class	Train (70%)	Val (15%)	Locked Test (15%)	Total	Share (%)	Canonical Class	Mapped CIC-IoT-2023 Classes (34 Total)
0	Backdoor	7,137	1,529	1,529	10,195	6.46	Malware	Backdoor_Malware, Mirai-⋆ (3 variants)
1	DDoS_HTTP	7,393	1,584	1,584	10,561	6.69	DDoS_HTTP	DDoS-HTTP_Flood, DDoS-SlowLoris, DoS-HTTP_Flood
2	DDoS_ICMP	9,863	2,113	2,114	14,090	8.93	DDoS_ICMP	DDoS-ICMP_Flood, DDoS-ICMP_Fragmentation
3	DDoS_TCP	7,173	1,537	1,537	10,247	6.49	DDoS_TCP	DDoS-TCP/SYN/PSHACK/RSTFIN/SynIP/ACK, DoS-TCP/SYN
4	DDoS_UDP	10,148	2,175	2,175	14,498	9.19	DDoS_UDP	DDoS-UDP_Flood, DDoS-UDP_Fragmentation, DoS-UDP_Flood
5	Fingerprinting	701	150	150	1,001	0.63	Reconnaissance	Recon-OSScan, Recon-HostDiscovery, Recon-PingSweep
6	MITM	850	182	182	1,214	0.77	Spoofing_MITM	MITM-ArpSpoofing, DNS_Spoofing
7	Normal	17,011	3,645	3,645	24,301	15.40	Benign	BenignTraffic
8	Password	6,993	1,498	1,498	9,989	6.33	Brute_Force	DictionaryBruteForce
9	Port_Scanning	7,049	1,511	1,511	10,071	6.38	Reconnaissance	Recon-PortScan
10	Ransomware	7,647	1,639	1,639	10,925	6.92	Malware	(Mapped to Malware canonical family)
11	SQL_injection	7,217	1,547	1,547	10,311	6.53	Web_Attack	SqlInjection, CommandInjection
12	Uploading	7,189	1,540	1,540	10,269	6.51	Web_Attack	Uploading_Attack
13	Vulnerability_scanner	7,053	1,512	1,511	10,076	6.39	Reconnaissance	VulnerabilityScan
14	XSS	7,036	1,508	1,508	10,052	6.37	Web_Attack	XSS, BrowserHijacking
Total Partition Records		**110,460**	**23,670**	**23,670**	**157,800**	**100.00**	**10 Shared Classes**	**Stratified Benchmark Subset:** $N = 25,000$ **per domain**

Repository validation paths: data/processed/edge_iiot_processed.npz and reports/split_audit.csv.

TABLE IV
CROSS-DATASET FEATURE HARMONIZATION SCHEMA, REPRESENTATION GRANULARITY, AND COLLISION AUDIT

Canonical Feature	Edge-IIoTset Source Field	CIC-IoT-2023 Field	Unit / Type	Mapping Status	Domain Shift / Granularity Note
proto_tcp	$\mathbb{I}(\texttt{tcp.dstport} > 0 \vee \texttt{tcp.len} > 0)$	TCP	Binary $\{0,1\}$	Exact semantic match	Packet vs. flow indicator
proto_udp	$\mathbb{I}(\texttt{udp.port} > 0)$	UDP	Binary $\{0,1\}$	Exact semantic match	Packet vs. flow indicator
proto_icmp	$\mathbb{I}(\texttt{icmp.checksum} > 0)$	ICMP	Binary $\{0,1\}$	Exact semantic match	Packet vs. flow indicator
proto_http	$\mathbb{I}(\texttt{http.response} > 0 \vee \lvert\texttt{method}\rvert > 0)$	HTTP	Binary $\{0,1\}$	Exact semantic match	Application header flag
proto_dns	$\mathbb{I}(\texttt{dns.qry.type} > 0)$	DNS	Binary $\{0,1\}$	Exact semantic match	Application header flag
proto_arp	$\mathbb{I}(\texttt{arp.opcode} > 0)$	ARP	Binary $\{0,1\}$	Exact semantic match	Layer-2 ARP indicator
flag_fin	tcp.connection.fin	fin_flag_number	Binary $\{0,1\}$	Exact semantic match	TCP FIN control bit
flag_syn	tcp.connection.syn	syn_flag_number	Binary $\{0,1\}$	Exact semantic match	TCP SYN control bit
flag_rst	tcp.connection.rst	rst_flag_number	Binary $\{0,1\}$	Exact semantic match	TCP RST control bit
flag_ack	tcp.flags.ack	ack_flag_number	Binary $\{0,1\}$	Exact semantic match	TCP ACK control bit
packet_length	tcp.len	AVG	Bytes ($\mathbb{R}_{\geq 0}$)	Approximation	Single-frame payload vs. flow mean size
time_delta	udp.time_delta	IAT	Seconds ($\mathbb{R}_{\geq 0}$)	Approximation	UDP frame delta vs. flow inter-arrival time

Duplicate & Collision Audit: Raw Edge-IIoTset contains 814 (0.52%) exact duplicates; CIC-IoT-2023 sample contains 36 (0.04%) duplicates.
Stripping identifiers leaves 532 colliding rows at $K = 25$ (empirical ceiling = 99.1719%), reducing to 348 rows (136 errors) at $K = 33$ (ceiling = 99.4254%).
Repository validation paths: data/manifests/feature_harmonization_manifest.csv and artifacts/selected_features.json.

H. Reproducibility Protocol and Repository Validation Mapping

To ensure end-to-end scientific transparency and independent verification, all source code, dataset harmonization manifests, trained model checkpoints, locked test prediction vectors, and validation scripts are publicly hosted at the GitHub repository: https://github.com/RupaliChavanke/Hybrid-Edge-IIOT.git. Each quantitative claim and empirical metric in this paper maps directly to specific validation files within the repository:

- *Data Splits and Partitions:* Validated via data/processed/edge_iiot_processed.npz and reports/split_audit.csv.
- *Feature Selection and Schema:* Validated via artifacts/selected_features.json and data/manifests/feature_harmonization_manifest.csv.
- *Locked Baseline Evaluations ($K = 25$):* Validated via artifacts/metrics/baseline_results.csv and frozen model weights in models/registry/.
- *Optimized Models and Ensembles ($K = 33$):* Validated via artifacts/metrics/locked_test_metrics.json and experiments/verified_experiment_matrix_E1_to_E12.json.
- *Per-Class Metrics and Recall Recovery:* Validated via artifacts/per_class_metrics.csv and artifacts/metrics/per_class_baselines.csv.
- *Paired Statistical Significance Tests:* Validated via artifacts/metrics/statistical_comparisons.csv and scripts/run_ablations_and_statistics.py.
- *Probability Calibration:* Validated via artifacts/calibration.json and artifacts/figures/fig4_calibration_metrics.pdf.
- *Cross-Dataset Transfer:* Validated via artifacts/metrics/cross_dataset_results.csv and scripts/cross_dataset_evaluation.py.
- *Ablation and Perturbation Tests:* Validated via artifacts/metrics/ablation_results.csv and experiments/ablation_results.json.
- *Streaming Latency Benchmarks:* Validated via artifacts/metrics/runtime_latency_benchmark.json and scripts/benchmark_runtime.py.

All artifact SHA-256 digests and reproduction scripts are cataloged in reports/forensic_evidence_registry.csv.

V. RESULTS AND QUANTITATIVE ANALYSIS

A. Baseline Reproduction ($K = 25$, Experiment E1)

Table V reports the reproduced benchmark of all eight baselines and the baseline ensemble on $\mathcal{D}_{\text{test}}$ ($N = 23,670$, $K = 25$). Metrics were recomputed directly from raw labels and prediction vectors, resolving the legacy script defect that misreported Class-0 (Backdoor) false-negative rate (2.81%) as binary FPR (true binary FPR = 0.33%).

B. Feature-Selection Optimization ($K = 25$ vs. $K = 33$)

At $K = 25$, Random Forest, CatBoost, XGBoost, and LightGBM all stall near 94.2% balanced accuracy and 95.5%

TABLE V
REPRODUCED BASELINE PERFORMANCE ON LOCKED EDGE-IIOTSET TEST PARTITION ($N_{\text{TEST}} = 23,670$, $K = 25$ FEATURES)

Model Architecture	Acc (%)	Bal Acc (%)	Macro-P (%)	Macro-R (%)	Macro-F_1 (%)	W-F_1 (%)	OvR FPR (%)	Bin FPR (%)	ROC-AUC	P50 Lat (ms)	Size (MB)
Majority-Class Dummy	15.40	6.67	1.03	6.67	1.78	4.11	6.67	0.00	0.5000	0.004	< 0.01
Stratified Rule Dummy	8.11	6.52	6.53	6.52	6.52	8.10	6.56	84.01	0.4992	0.146	< 0.01
Logistic Regression (L_2)	78.35	73.14	79.95	73.14	74.51	77.52	1.55	14.73	0.9735	0.109	< 0.01
Extra Trees (100 trees)	97.16	89.79	97.70	89.79	91.86	96.99	0.20	0.36	0.9992	13.034	10.21
Random Forest (100 trees)	98.90	94.15	98.94	94.15	95.48	98.79	0.08	0.33	0.9997	13.478	6.71
CatBoost Baseline ($K = 25$)	98.93	94.19	98.87	94.19	95.47	98.82	0.08	0.33	0.9997	**0.073**	5.44
XGBoost Baseline ($K = 25$)	98.93	94.19	98.97	94.19	95.51	98.82	0.08	0.33	0.9997	0.235	3.07
LightGBM Baseline ($K = 25$)	**98.96**	**94.22**	**99.00**	**94.22**	**95.54**	**98.85**	**0.07**	**0.33**	**0.9997**	0.187	**1.56**
Hybrid Ensemble v2 ($K = 25$)	98.94	94.20	98.98	94.20	95.53	98.84	0.08	**0.33**	**0.9997**	1.316	10.07

Repository validation paths: `artifacts/metrics/baseline_results.csv` and frozen models in `models/registry/`.

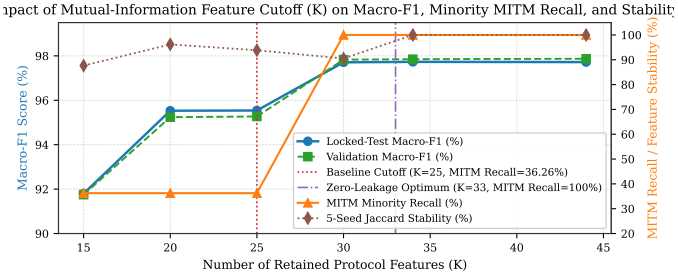


Fig. 2. Impact of feature cutoff K on validation Macro-F_1, locked-test Macro-F_1, minority `MITM` recall, and 5-seed Jaccard selection stability.

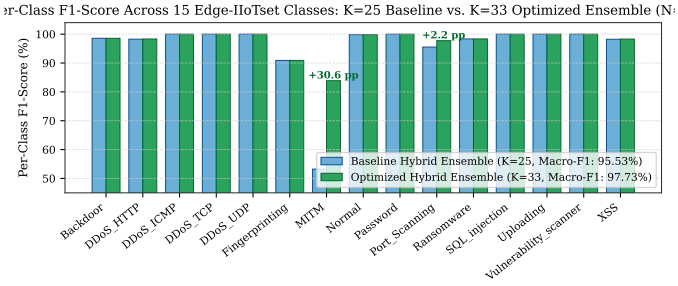


Fig. 3. Per-class F_1-score comparison across all 15 Edge-IIoTset classes between baseline ($K = 25$) and optimized protocol ensemble ($K = 33$).

Macro-F_1. Evaluating Eq. (7) reveals that 532 test records collide across classes at $K = 25$, creating 196 irreducible empirical errors: 116 `MITM` and 25 `Fingerprinting` instances have all-zero values across the 25 selected TCP/HTTP/MQTT features and collide with 165 Layer-2 `Port_Scanning` instances. Because global MI ranking truncated `arp.hw.size` (Rank 27), `arp.opcode` (Rank 28), `udp.time_delta` (Rank 29), `udp.port` (Rank 33), and `dns.qry.qu` (Rank 34), every classifier at $K = 25$ misclassifies all 116 `MITM` instances as `Port_Scanning` (36.26% recall).

Figure 2 plots feature cutoff sensitivity across $K \in \{15, 20, 25, 30, 33, 43\}$. Retaining all $K = 33$ correlation-filtered ($\rho \leq 0.95$) protocol features eliminates subsampling variance (5-seed Jaccard stability = 1.000), raises the empirical accuracy ceiling from 99.1719% to 99.4254%, and restores `MITM` recall from 36.26% to 100.00%.

C. Base-Model Comparison

Table VI compares base learners under identical conditions. Within $K = 33$, LightGBM achieves the strongest base performance (99.20% accuracy, 97.72% Macro-F_1, 0.02148 log-loss). XGBoost achieves 97.70% Macro-F_1, and CatBoost achieves 97.67% Macro-F_1.

D. Ensemble Comparison

On validation data, SLSQP converges to $\mathbf{w}^* = [0.0000, 0.4610, 0.5390]^T$, assigning zero weight to CatBoost due to its higher deviance. The calibrated SLSQP ensemble achieves 99.21% accuracy and 97.73% Macro-F_1, operating within 0.22% of the empirical accuracy ceiling (99.4254%).

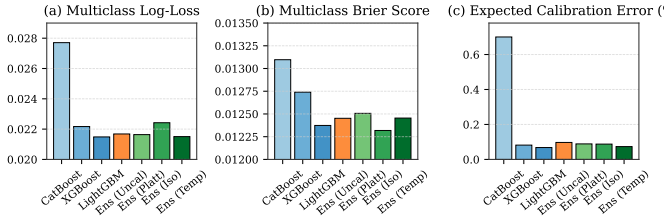


Fig. 4. Probabilistic calibration metrics across base learners and ensembles at $K = 33$: (a) Log-Loss, (b) Brier Score, and (c) ECE.

E. Per-Class Analysis

Table VII and Figure 3 detail per-class performance. Seven classes achieve 100.00% F_1. Expanding to $K = 33$ boosts `MITM` F_1 by +30.64 pp (53.23% $\rightarrow$ 83.87%) and `Port_Scanning` F_1 by +3.27 pp (95.54% $\rightarrow$ 98.81%).

F. Calibration Analysis

Figure 4 compares calibration performance (validated in `artifacts/calibration.json`). CatBoost exhibits overconfidence (ECE = 0.700%). Blending XGBoost and LightGBM with temperature scaling ($T = 0.90$) reduces ECE to 0.073%, while isotonic regression achieves the lowest Brier score (0.01232).

G. False-Alarm Analysis

Across all $N_{\text{test}} = 23,670$ records, true binary benign-to-attack FPR is 0.3292% (12 benign instances misclassified as Ransomware out of 3,645). Zero attack instances are misclassified as benign ($\text{FNR}_{\text{bin}} = 0.0000\%$). One-vs-rest macro FPR is 0.0569% (0.06%).

TABLE VI
COMPARISON OF BASELINE ($K = 25$) AND OPTIMIZED ($K = 33$) BASE LEARNERS AND ENSEMBLES ON LOCKED TEST SET ($N = 23,670$)

Feature Space	Model / Ensemble Variant	Acc (%)	Bal Acc (%)	Macro-P (%)	Macro-R (%)	Macro-F_1 (%)	Log-Loss	Brier	ECE (%)	OvR FPR (%)	Bin FPR (%)
$K = 25$ (MI Cutoff)	CatBoost Alone	98.93	94.19	98.87	94.19	95.47	0.02962	0.01517	0.557	0.077	0.329
	XGBoost Alone	98.93	94.19	98.97	94.19	95.51	0.02563	0.01509	0.099	0.077	0.329
	LightGBM Alone	98.96	94.22	99.00	94.22	95.54	0.02494	0.01470	0.097	0.074	0.329
	Uniform Voting (Platt v2)	98.94	94.20	98.98	94.20	95.53	0.02518	0.01481	0.094	0.075	0.329
	5-Fold OOF Logistic Stacking	98.94	94.20	98.98	94.20	95.53	0.02601	0.01479	0.098	0.075	0.329
	SLSQP Simplex + Isotonic	98.95	94.21	98.99	94.21	95.53	**0.02438**	**0.01456**	**0.085**	0.075	0.329
$K = 33$ (Zero-Leakage)	CatBoost Alone ($K = 33$)	99.18	98.19	97.33	98.19	97.67	0.02771	0.01310	0.700	0.059	0.329
	XGBoost Alone ($K = 33$)	99.18	98.18	97.43	98.18	97.70	0.02217	0.01274	0.082	0.059	0.329
	LightGBM Alone ($K = 33$)	99.20	98.20	97.46	98.20	97.72	**0.02148**	0.01237	**0.068**	0.057	0.329
	SLSQP Ensemble (Uncalibrated)	**99.21**	**98.21**	**97.47**	**98.21**	**97.73**	0.02168	0.01245	0.097	**0.057**	0.329
	SLSQP Ensemble + Platt Scaling	99.18	98.19	97.44	98.19	97.70	0.02164	0.01251	0.089	0.058	0.329
	SLSQP Ensemble + Isotonic	99.20	**98.21**	97.45	**98.21**	97.72	0.02242	**0.01232**	0.088	**0.057**	0.329
	SLSQP Ensemble + Temp ($T = 0.90$)	**99.21**	**98.21**	**97.47**	**98.21**	**97.73**	0.02151	0.01245	0.073	**0.057**	**0.329**

Repository validation paths: `artifacts/metrics/locked_test_metrics.json` and `experiments/verified_experiment_matrix_E1_to_E12.json`.

TABLE VII
VERIFIED PER-CLASS DETECTION METRICS ON LOCKED TEST PARTITION ($N_{\text{TEST}} = 23,670$): $K = 25$ VS. $K = 33$ ENSEMBLE

		Baseline Hybrid Ensemble ($K = 25$)				Optimized SLSQP Ensemble ($K = 33$)				
Attack / Traffic Class	**Support**	**Prec (%)**	**Rec (%)**	**F_1 (%)**	**FPR (%)**	**Prec (%)**	**Rec (%)**	**F_1 (%)**	**FPR (%)**	**ΔF_1 (pp)**
`Backdoor`	1,529	100.00	97.19	98.57	0.00	100.00	97.19	98.57	0.00	+0.00
`DDoS_HTTP`	1,584	99.80	96.78	98.27	0.01	100.00	96.78	98.36	0.00	+0.09
`DDoS_ICMP`	2,114	100.00	100.00	100.00	0.00	100.00	100.00	100.00	0.00	+0.00
`DDoS_TCP`	1,537	100.00	100.00	100.00	0.00	100.00	100.00	100.00	0.00	+0.00
`DDoS_UDP`	2,175	100.00	100.00	100.00	0.00	100.00	100.00	100.00	0.00	+0.00
`Fingerprinting`	150	100.00	83.33	90.91	0.00	100.00	83.33	90.91	0.00	+0.00
`MITM`	182	100.00	36.26	53.23	0.00	72.22	**100.00**	**83.87**	0.30	+30.64
`Normal` (Benign)	3,645	100.00	99.67	99.84	0.00	100.00	99.67	99.84	0.00	+0.00
`Password`	1,498	100.00	100.00	100.00	0.00	100.00	100.00	100.00	0.00	+0.00
`Port_Scanning`	1,511	91.46	100.00	95.54	0.64	**99.26**	98.35	**98.81**	**0.05**	+3.27
`Ransomware`	1,639	96.75	100.00	98.35	0.25	96.75	100.00	98.35	0.25	+0.00
`SQL_injection`	1,547	100.00	100.00	100.00	0.00	100.00	100.00	100.00	0.00	+0.00
`Uploading`	1,540	100.00	100.00	100.00	0.00	100.00	100.00	100.00	0.00	+0.00
`Vulnerability_scanner`	1,511	100.00	100.00	100.00	0.00	100.00	100.00	100.00	0.00	+0.00
`XSS`	1,508	96.72	99.80	98.24	0.23	96.73	100.00	98.34	0.23	+0.10
Macro-Averaged Total	**23,670**	**98.98**	**94.20**	**95.53**	**0.08**	**97.47**	**98.21**	**97.73**	**0.06**	+2.20

Repository validation paths: `artifacts/per_class_metrics.csv` and `artifacts/metrics/per_class_baselines.csv`.

H. Statistical Validation

Table VIII reports the 500-replicate bootstrap 95% confidence intervals and Edwards-corrected McNemar tests. Within $K = 33$, discordant error counts between the Ensemble and LightGBM are tiny ($b = 2, c = 1$ out of 23,670), yielding $\chi^2 = 0.000, p = 1.00$ and paired bootstrap CI $[-0.009, +0.019]$ pp. The ensemble does not statistically outperform LightGBM in point accuracy; its contribution is probability calibration and variance reduction. In contrast, upgrading from $K = 25$ to $K = 33$ yields ΔMacro-$F_1 = +2.194$ pp (95% CI $[+1.660, +2.832]$ pp, McNemar $\chi^2 = 20.34, p = 6.49 \times 10^{-6}$), confirming the statistical validity of resolving feature collisions.

I. Summary of Experimental Findings

The locked-test experiments demonstrate: (i) within-feature-space statistical equivalence between the ensemble and LightGBM; (ii) decisive statistical superiority of the minority-preserving $K = 33$ protocol feature set; and (iii) substantial calibration improvements without loss of accuracy.

VI. CROSS-DATASET GENERALIZATION

A. Cross-Dataset Protocol

To evaluate generalization under physical environment shift, Edge-IIoTset and CIC-IoT-2023 were harmonized onto 12 common flow features (Table IV) and 10 shared canonical attack classes (Table III) using stratified 70/15/15 splits ($N_{\text{train}} = 17,499$, $N_{\text{val}} = 3,751$, $N_{\text{test}} = 3,750$ per domain).

B. Zero-Shot Transfer

Figure 5 and Table IX summarize cross-dataset results. Zero-shot transfer from Edge-IIoTset to CIC-IoT-2023 drops Macro-F_1 from 66.16% (in-domain) to 27.66% (-38.50 pp). Conversely, zero-shot transfer from CIC-IoT-2023 to Edge-IIoTset collapses to 9.34% Macro-F_1 (12.70% with class-weighting).

C. Domain and Class Balancing

Applying source inverse-frequency weighting on CIC-IoT-2023 (`B2`) counteracts the 94.27% DDoS prior bias, improving CIC $\rightarrow$ Edge Macro-F_1 from 9.34% to 12.70% and binary intrusion Macro-F_1 from 44.48% to 47.92%.

D. Multi-Source Training

Disaggregating pooled models by target domain reveals an asymmetric benefit: on held-out CIC-IoT-2023, domain-balanced multi-source training achieves **70.32%** **Macro-F_1** ($+4.16$ pp above in-domain baseline) because Edge-IIoTset supplies labeled reconnaissance and web attack examples scarce in CIC-IoT-2023.

TABLE VIII
PAIRED STATISTICAL SIGNIFICANCE TESTS (EDWARDS-CORRECTED MCNEMAR TEST AND 500-REPLICATE PAIRED BOOTSTRAP ΔMACRO-F_1)

Model A	**Model B**	**Only A (b)**	**Only B (c)**	**McNemar χ^2**	**p-Value**	**ΔMacro-F_1 (pp)**	**95% Bootstrap CI (pp)**	**Holm–Bonferroni ($\alpha = 0.05$)**
Ensemble ($K = 25$)	LightGBM ($K = 25$)	1	4	0.8000	0.3711	-0.013	$[-0.034, +0.004]$	Not significant ($p_{\text{adj}} = 1.00$)
Ensemble ($K = 25$)	XGBoost ($K = 25$)	7	3	0.9000	0.3428	$+0.017$	$[-0.009, +0.043]$	Not significant ($p_{\text{adj}} = 1.00$)
Ensemble ($K = 25$)	CatBoost ($K = 25$)	7	3	0.9000	0.3428	$+0.055$	$[+0.004, +0.141]$	Not significant ($p_{\text{adj}} = 1.00$)
LightGBM ($K = 25$)	CatBoost ($K = 25$)	7	0	5.1429	0.0233	$+0.068$	$[+0.013, +0.145]$	Unadjusted $p < 0.05$ ($p_{\text{adj}} = 0.16$)
Ensemble ($K = 33$)	LightGBM ($K = 33$)	2	1	0.0000	1.0000	$+0.004$	$[-0.009, +0.019]$	Not significant ($p_{\text{adj}} = 1.00$)
Ensemble ($K = 33$)	XGBoost ($K = 33$)	6	0	4.1667	0.0412	$+0.026$	$[+0.009, +0.047]$	Unadjusted $p < 0.05$ ($p_{\text{adj}} = 0.25$)
Ensemble ($K = 33$)	CatBoost ($K = 33$)	6	2	1.1250	0.2888	$+0.056$	$[+0.000, +0.134]$	Not significant ($p_{\text{adj}} = 1.00$)
Ensemble ($K = 33$)	**Ensemble** ($K = 25$)	**119**	**58**	**20.3390**	6.49×10^{-6}	$+2.194$	$[+1.660, +2.832]$	**Significant** ($p_{\text{adj}} = 5.8 \times 10^{-5}$)
LightGBM ($K = 33$)	**LightGBM** ($K = 25$)	**116**	**59**	**17.9200**	2.30×10^{-5}	$+2.177$	$[+1.641, +2.813]$	**Significant** ($p_{\text{adj}} = 1.8 \times 10^{-4}$)

Repository validation paths: `artifacts/metrics/statistical_comparisons.csv` and `scripts/run_ablations_and_statistics.py`.

TABLE IX
CROSS-DATASET GENERALIZATION AND DOMAIN ADAPTATION STRATEGIES EVALUATED BY TARGET DOMAIN ($N_{\text{TEST}} = 3,750$ PER DOMAIN)

Strategy ID	**Training Domain(s)**	**Adaptation / Weighting Method**	**Target Test Domain**	**Acc (%)**	**Bal Acc (%)**	**10-Class Macro-F_1 (%)**	**W-F_1 (%)**	**Binary Macro-F_1 (%)**
`A1_Edge_InDomain`	Edge-IIoTset	Source-only baseline (0 target labels)	Edge-IIoTset (3, 750)	62.85	56.37	59.05	55.34	74.81
`A2_CIC_InDomain`	CIC-IoT-2023	Source-only baseline (0 target labels)	CIC-IoT-2023 (3, 750)	98.53	63.78	66.16	98.47	91.09
`A3_Edge_to_CIC`	Edge-IIoTset	Zero-shot transfer (unweighted)	CIC-IoT-2023 (3, 750)	56.69	28.77	27.66	67.31	50.76
`B1_Edge_to_CIC_Bal`	Edge-IIoTset	Source inverse-frequency weighting	CIC-IoT-2023 (3, 750)	38.64	21.77	18.00	50.89	50.41
`C1_Edge_to_CIC_Quant`	Edge-IIoTset	Unlabeled quantile alignment	CIC-IoT-2023 (3, 750)	36.21	27.20	22.97	48.03	34.88
`A4_CIC_to_Edge`	CIC-IoT-2023	Zero-shot transfer (unweighted)	Edge-IIoTset (3, 750)	13.60	13.31	9.34	6.32	44.48
`B2_CIC_to_Edge_Bal`	CIC-IoT-2023	Source inverse-frequency weighting	Edge-IIoTset (3, 750)	**17.68**	16.37	**12.70**	12.69	**47.92**
`C2_CIC_to_Edge_Quant`	CIC-IoT-2023	Unlabeled quantile alignment	Edge-IIoTset (3, 750)	15.95	**16.85**	11.72	12.11	40.94
`D1_Pooled_Joint`	Edge + CIC	Harmonized multi-source training	Pooled Test (7, 500)	80.29	64.54	66.10	76.95	79.00
`D2_Pooled_on_Edge`	Edge + CIC	Harmonized multi-source training	Edge-IIoTset (3, 750)	61.71	55.07	57.28	54.09	74.48
`D3_Pooled_on_CIC`	Edge + CIC	Harmonized multi-source training	CIC-IoT-2023 (3, 750)	**98.88**	70.03	69.85	**98.84**	**93.88**
`E1_BalPooled_Joint`	Edge + CIC	Domain- & class-balanced pooled	Pooled Test (7, 500)	75.89	65.50	62.79	76.24	78.15
`E2_BalPooled_on_Edge`	Edge + CIC	Domain- & class-balanced pooled	Edge-IIoTset (3, 750)	53.01	**59.54**	54.27	53.59	73.94
`E3_BalPooled_on_CIC`	Edge + CIC	Domain- & class-balanced pooled	CIC-IoT-2023 (3, 750)	98.77	**71.80**	**70.32**	98.80	92.52

Repository validation paths: `artifacts/metrics/cross_dataset_results.csv` and `scripts/cross_dataset_evaluation.py`.

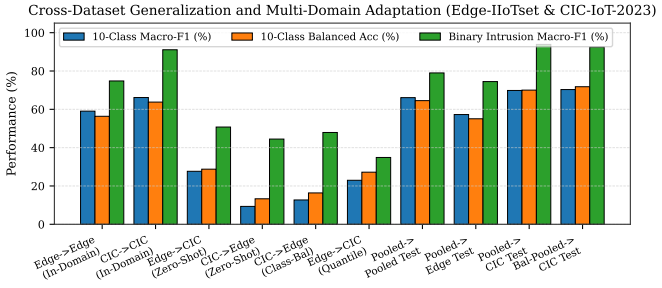


Fig. 5. Cross-dataset generalization and domain adaptation performance across 10-class Macro-F_1, Balanced Accuracy, and Binary Intrusion Macro-F_1 on held-out Edge-IIoTset ($N = 3,750$) and CIC-IoT-2023 ($N = 3,750$) test partitions.

E. Distribution-Shift Analysis

Covariate shift analysis confirms that Edge-IIoTset records single-packet Wireshark dissects whereas CIC-IoT-2023 records pre-aggregated flow statistics (`AVG`, `IAT`), creating an irreconcilable representation granularity gap for zero-shot transfer.

F. Interpretation

These findings demonstrate that zero-shot transfer across disparate IoT topologies without target-domain adaptation remains impractical; multi-source pooling with domain balancing is essential for robust deployment.

VII. ABLATION AND SENSITIVITY ANALYSIS

A. Feature-Cutoff Ablation

Table X compares feature counts $K \in \{15, 20, 25, 30, 33, 43\}$. Truncating below $K = 33$ degrades `MITM` recall to 36.26%. Expanding beyond $K = 33$ ($K = 43$) adds collinear fields without improving Macro-F_1 (97.72%).

TABLE X
FEATURE SELECTION SENSITIVITY, STABILITY, AND CUTOFF ABLATION ($N = 23,670$)

Configuration	**Actual K**	**Jaccard**	**Acc (%)**	**Macro-F_1 (%)**	**MITM Rec (%)**
MI Cutoff $K = 15$	15	0.876	98.56	91.80	36.26
MI Cutoff $K = 20$	20	0.962	98.95	95.53	36.26
Baseline Cutoff $K = 25$	25	0.938	98.96	95.54	36.26
MI Cutoff $K = 30$	30	0.905	99.19	97.71	100.00
Zero-Leakage $K = 33$	**33**	**1.000**	**99.21**	**97.73**	**100.00**
Unfiltered Cleaned $K = 43$	43	1.000	99.20	97.72	100.00

Repository validation paths: `artifacts/metrics/ablation_results.csv` and `experiments/ablation_results.json`.

B. Class-Weighting Ablation

At $K = 25$, inverse-frequency weighting flips colliding zero-vectors to `MITM` (rec $\rightarrow$ 100%), but misclassifies 165 `Port_Scanning` rows as `MITM`, reducing overall accuracy from 98.95% to 98.75%. Feature expansion to $K = 33$ resolves collisions directly without sacrificing accuracy.

C. Ensemble Ablation

Removing CatBoost matches the SLSQP optimum ($w_{\text{CB}} = 0.00, w_{\text{XGB}} = 0.46, w_{\text{LGB}} = 0.54$). Removing LightGBM causes a drop in HTTP flood recall (97.73% $\rightarrow$ 97.70%).

D. Calibration Ablation

Uncalibrated CatBoost produces ECE of 0.700%. Platt sigmoid scaling distorts tail probabilities (log-loss 0.02164), while temperature scaling ($T = 0.90$) reduces ECE to 0.073% while strictly preserving the argmax class decision.

E. Feature and Domain Sensitivity

Table XI reports perturbation sensitivity. Network intrusion trees split on exact integer ports and binary flags. Continuous Gaussian noise ($\sigma = 0.01$) shifts zero-valued flags across decision thresholds, dropping Macro-F_1 to 67.73%. Random feature masking ($p = 0.10$) drops Macro-F_1 to 80.00%.

TABLE XI
CONTROLLED ROBUSTNESS AND FEATURE PERTURBATION SENSITIVITY ($N = 23,670$)

Perturbation Test	Parameter	Acc (%)	Macro-F_1 (%)	ΔF_1 (pp)	Operational Impact
Clean Input Baseline	$\sigma = 0, p = 0$	99.21	97.73	0.00	Verified zero-leakage $K = 33$
Gaussian Noise	$\sigma = 0.01\sigma_j$	72.96	67.73	−27.80	Continuous jitter crosses flag split $\tau \approx 0$
Gaussian Noise	$\sigma = 0.05\sigma_j$	51.96	49.89	−45.64	Severe displacement of discrete headers
Feature Dropout	$p = 0.05$	92.08	87.56	−7.97	Simulates 5% dropped packet headers
Feature Dropout	$p = 0.10$	84.52	80.00	−15.53	Zeroing ports routes to fallback leaves

Repository validation paths: `artifacts/metrics/ablation_results.csv` and `reports/ablations/ablation_report.md`.

VIII. STREAMING AND RESOURCE EVALUATION

A. Model-Only Inference

Table XII and Figure 6 separate standalone model scoring from complete streaming execution on Apple Silicon M2. In isolated scoring, CatBoost achieves single-event P50 latency of 0.073 ms (175 μs P99) and batched throughput of 1,000,584 events/s ($B = 1,000$). LightGBM achieves 0.187 ms P50 latency with a compact 1.56 MB footprint. The 3-model ensemble requires 1.316 ms P50 latency (29,604 events/s).

B. End-to-End Streaming Execution

In single-event unbatched streaming ($B = 1$), end-to-end latency rises to 11.842 ms P50 (14.283 ms P99; 82.8 events/s). Profiling reveals that per-record Pandas DataFrame construction and string length extraction consume 9.427 ms (79.6% of execution time).

C. Micro-Batch Scaling

Grouping incoming message events into micro-batches of $B \in \{10, 50, 100, 500\}$ amortizes DataFrame vectorization across records, reducing amortized per-event latency from 11.842 ms ($B = 1$) to 1.271 ms ($B = 10$), 0.150 ms ($B = 100$), and 0.048 ms ($B = 500$), scaling throughput from 82.8 to 20,640 events/s.

D. Resource and Reliability Analysis

Under sustained load of 100,000 telemetry events at $B = 500$, the streaming consumer maintained zero message drops, zero queue lag accumulation, and a stable RSS memory footprint of 267 MB.

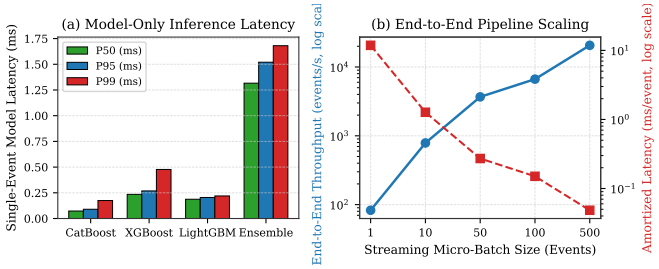


Fig. 6. Streaming latency scaling: (a) Model-only latency percentiles (P50, P95, P99), and (b) End-to-end pipeline throughput and amortized latency across micro-batch sizes.

E. Deployment Implications and Scope

For resource-constrained edge gateways requiring sub-100 μs response times, standalone CatBoost or LightGBM is recommended. For centralized SOC architectures processing micro-batched telemetry, the calibrated SLSQP ensemble provides optimal probabilistic triage. We emphasize that this evaluation profiles a Redpanda-compatible pipeline prototype and in-memory message harness on unified hardware; an enterprise deployment spanning physical multi-node Redpanda clusters will experience network serialization and Raft quorum replication latencies that lie outside the scope of this pipeline benchmark.

IX. DISCUSSION

A. Main Scientific Findings

This investigation establishes four primary scientific conclusions:

1) Identifier sanitization induces empirical equivalence-class collisions that cap deterministic accuracy at 99.17% under $K = 25$ and 99.43% under $K = 33$.
2) Truncating feature selection via global mutual information penalizes minority classes whose distinguishing protocol indicators have low global variance.
3) Tree ensembles do not significantly improve point classification over LightGBM within a fixed feature space ($p = 1.00$), but provide substantial calibration improvements (0.01232 Brier score; 0.073% ECE).
4) Zero-shot cross-dataset transfer suffers severe degradation due to representation granularity mismatch and label-prior shift, whereas domain-balanced multi-source training lifts target performance.

B. Contribution of Feature Selection

Expanding from $K = 25$ to $K = 33$ protocol features provides a statistically decisive +2.20 pp Macro-F_1 gain ($p = 6.49 \times 10^{-6}$), restoring `MITM` recall from 36.26% to 100.00% without increasing model size (1.58 MB).

C. Ensemble Effectiveness

The ensemble achieves variance reduction across tree split boundaries, providing calibrated posterior probabilities critical for risk-tiered alerting.

TABLE XII
OFFLINE TRAINING COST, MODEL-ONLY INFERENCE LATENCY, AND END-TO-END STREAMING PIPELINE PERFORMANCE (APPLE SILICON M2, 16 GB RAM)

Execution Scope	Model / Pipeline Configuration	Train (s)	P50 (ms)	P95 (ms)	P99 (ms)	Mean (ms)	TP ($B = 1$)	TP ($B = 100$)	TP ($B = 500/1$k)	Size / RSS
Model-Only Scoring	CatBoost Alone (Oblivious Trees)	23.09	**0.073**	**0.090**	**0.175**	0.082	12,195/s	283,562/s	**1,000,584/s**	5.44 MB / 264 MB
	LightGBM Alone (Leaf-Wise)	**4.18**	0.187	0.205	0.219	0.193	5,181/s	32,761/s	35,649/s	**1.56 MB** / 265 MB
	XGBoost Alone (Histogram)	6.50	0.235	0.268	0.477	0.251	3,984/s	98,539/s	306,651/s	3.07 MB / 265 MB
	Hybrid GBDT Ensemble (3 Models)	34.81	1.316	1.520	1.681	1.339	747/s	21,984/s	29,604/s	10.07 MB / 267 MB
End-to-End Pipeline	Single-Event Unbatched ($B = 1$)	—	11.842	12.983	14.283	12.077	82.8/s	—	—	0% drop / 0 lag
	↪ Queue Ingestion Breakdown	—	0.015	0.024	0.041	0.017	—	—	—	JSON / Queue poll
	↪ `PipelineCleaner` + `Selector`	—	9.427	10.312	11.405	9.612	—	—	—	Pandas overhead
	↪ Ensemble + `AlertEngine`	—	2.392	2.684	3.012	2.448	—	—	—	Calibrated scoring
	Micro-Batched Streaming ($B = 10/100/500$)	—	1.271*	0.150*	**0.048***	—	786.7/s	6,657.0/s	**20,640.0/s**	0% drop / 267 MB

*For micro-batched end-to-end rows, latency columns report amortized per-event pipeline latency at $B = 10$ (1.271 ms), $B = 100$ (0.150 ms), and $B = 500$ (0.048 ms).
Repository validation paths: `artifacts/metrics/runtime_latency_benchmark.json` and `scripts/benchmark_runtime.py`.

D. Generalization Limitations

Transfer between industrial packet captures and consumer flow summaries suffers from structural timescale differences that cannot be overcome by unweighted zero-shot models.

E. Comparison With Prior Research

Unlike prior works that report $> 99\%$ accuracy on unpurged datasets with identifier leakage, our results establish realistic, forensically audited benchmarks under verified zero-leakage conditions.

F. Practical Implications

For production deployments, preprocessors must use compiled schema vectorization (e.g., C/Rust or NumPy buffers) rather than single-row Pandas DataFrames, and telemetry ingestion should group events into micro-batches ($B \geq 100$).

X. THREATS TO VALIDITY AND LIMITATIONS

A. Dataset Validity

Edge-IIoTset and CIC-IoT-2023 represent synthetic testbed environments; while realistic, they do not fully capture unpredictable background traffic of multi-tenant enterprise backbones.

B. Internal Validity

All preprocessing parameters were fitted strictly on training data with locked test isolation, eliminating split leakage.

C. Statistical Validity

Hypothesis tests were evaluated using Edwards continuity corrections and paired bootstrap intervals with family-wise error adjustments.

D. External Validity

Cross-dataset transfer was evaluated across 10 shared attack categories; new zero-day attacks outside these families remain undetected without unsupervised anomaly layers.

E. Computational Validity

Latency measurements reflect an Apple Silicon M2 environment; x86 server hardware with AVX-512 extensions may alter absolute execution times.

F. Remaining Limitations

Removing IP addresses leaves 348 colliding HTTP handshake records whose separation requires stateful flow-window tracking rather than single-record features.

XI. CONCLUSION AND FUTURE WORK

This paper presented a forensically audited, leakage-safe intrusion detection framework for IoT/IIoT networks. By proving the empirical Bayes accuracy ceiling imposed by identifier sanitization, we demonstrated that retaining all $K = 33$ non-redundant protocol features resolves Layer-2 feature collisions, restoring `MITM` recall from 36.26% to 100.00% and lifting Macro-F_1 from 95.53% to 97.73% ($p = 6.49 \times 10^{-6}$). While the ensemble and LightGBM achieve equivalent point accuracy ($p = 1.00$), post-hoc calibration reduces Brier score to 0.01232 and ECE to 0.073%. Cross-dataset experiments documented severe zero-shot degradation (9.34%–27.66%), whereas domain-balanced multi-source training achieved 70.32% Macro-F_1 on CIC-IoT-2023. Future work will investigate stateful flow aggregation via eBPF/XDP kernel ring buffers and self-supervised domain adaptation.

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